

Ching-Yi Lin, PhD

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8223 Paint Branch Dr., College Park, MD 20740

Updated January 18, 2026

EDUCATION

Ph.D. in Electrical and Computer Engineering

August 2018 - January 2025

Carnegie Mellon University, Pittsburgh, PA

Dissertation title: A Multi-Modal Computational Framework for in vitro Cancer Cell Analysis

Based on Capacitance Sensing and Optical Imaging

Advisor: Marc Dandin, Daniel Bankman (2019-2020), and Radu Marculescu (2018-2019)

Eta Kappa Nu (HKN) Society

M.S. in Electrical and Computer Engineering

August 2018 - December 2022

Carnegie Mellon University, Pittsburgh, PA

B.S. in Electrical Engineering

August 2013 - June 2017

National Tsing Hua University (NTHU), Hsinchu, Taiwan

Class Representative

RESEARCH INTERESTS

Energy-efficient circuits, nonlinear model training, biosensor design, and machine learning accelerators.

WORK EXPERIENCE

Postdoctoral Associate

March 2025 - Present

University of Maryland, College Park

- Advisor: Sahil Shah and Pamela A. Abshire
- Developed 3-bit model integerization algorithm to reduce the precision required during the inference stage
- Realized 3-bit vision transformer on systolic array using FPGA targeting energy-efficient ML inference.
- Trained model considering non-linearity and process variations in resistive RAM (ReRAM)
- Co-lectured *Embedded Machine Learning*

Digital Circuit Design Intern

Summer 2021

Apple Inc.

- Manager: Amrinder Barn
- Designed and evaluated customized SRAM cells at schematic level
- Modeled and optimized high-speed transmission lines

Software Engineer Intern

Summer 2016

Industrial Technology Research Institute (ITRI), Taiwan

- Research group: Information and communications research laboratories
- Manager: Li-Han Chen
- Installed Android 5.1 on Odroid-C2, a 64-bit quad-core SBC
- Ported Secure Virtual Mobile Platform (SVMP) on Android 5.1

HONORS AND AWARDS

CMLH Translational Fellowship

2024

Proposal topics: Towards Long-term Data-driven Assays with Non-Visual Sensors
Sponsored by the Center for Machine Learning and Health (CMLH)

IEEE Circuits and Systems Society Student Travel Grants

2024

Topics: Towards a Lab-on-a-Chip as a Service (LoCaaS) Framework for *in Vitro* Cancer Cell Assays

Carnegie Institute of Technology Dean's Fellow

2018

TEACHING EXPERIENCE

Co-instructor, Embedded Machine Learning

Fall 2025

University of Maryland, College Park

- Graduate-level course co-taught with Prof. Sahil Shah
- Topics: Transformer models, model quantization/integerization, switched capacitor arrays.

Teaching Assistant

Carnegie Mellon University

- Analog Integrated Circuit Design (Prof. Marc Dandin)
- Intro to Machine Learning for Engineers (Prof. Carlee Joe-Wong)

Fall 2023

Spring 2022

Teaching Assistant

NTHU

- Embedded System Lab (Prof. Jing-Jia Liou)
- Advanced Computer Architecture (Prof. Yarsun Hsu)
- Introduction to C Programming (Prof. Mi-Chang Chang)

Spring 2017 & 2018

Fall 2017

Fall 2016

BOOK AND BOOK CHAPTERS

- [B1] **Ching-Yi Lin**, Md Sakibur Sajal, Yann Gilpin, Fahimeh Dehghandehnavi, Anna Batueva, Kai-Chun Lin, Nicole McFarlane, and Marc Dandin. "CMOS bioelectronics: Current and future trends." *Bioelectronics* (2022): 93-107.

JOURNAL PAPERS

- [J7] Yongmin Baek, Byungjoon Bae, Hyojin Shin, Charaana Sonnadara, Haein Cho, **Ching-Yi Lin**, Yujia Mu, Cong Shen, Sahil Shah, Gunuk Wang, and Kyusang Lee. "Edge intelligence through in-sensor and near-sensor computing for the artificial intelligence of things." , 2(1), p.25 (2025).
- [J6] Yann Gilpin, **Ching-Yi Lin**, and Marc Dandin. "Evaporation and focus degradation mitigation in in-incubator live cell imaging for capacitance lab-on-CMOS microsystem calibration." *Review of Scientific Instruments*, 96(9) (2025).
- [J5] **Ching-Yi Lin** and Marc Dandin. "Machine Learning Identification and Classification of Cancer Cell Behaviors in a Lab-on-CMOS Capacitance Sensing Platform." *IEEE Journal of Biomedical and Health Informatics*, doi: 10.1109/JBHI.2024.3486251 (2024).
- [J4] Kyle Smith, **Ching-Yi Lin**, Yann Gilpin, Elizabeth Wayne, and Marc Dandin. "Measuring and modeling macrophage proliferation in a lab-on-CMOS capacitance sensing microsystem." *Frontiers in Bioengineering and Biotechnology*, 11: 1159004 (2023).
- [J3] Yuwei Qin, Ruben Purdy, Alec Probst, **Ching-Yi Lin**, and Jian-Gang Zhu. "Non-Linear CNN-Based Read Channel for Hard Disk Drive With 30% Error Rate Reduction and Sequential 200-Mbits/s Throughput in 28-nm CMOS." *IEEE Journal of Solid-State Circuits*, 58, no. 4: 1094-1105 (2023).

- [J2] Yuwei Qin, Ruben Purdy, Alec Probst, **Ching-Yi Lin**, and Jian-Gang Zhu. "ASIC Implementation of Nonlinear CNN-Based Data Detector for TDMR System in 28 nm CMOS at 200 Mbits/s Throughput." IEEE Transactions on Magnetics, 59, no. 3: 1-8 (2022).
- [J1] Kartikeya Bhardwaj, **Ching-Yi Lin**, Anderson Sartor, and Radu Marculescu. "Memory- and communication-aware model compression for distributed deep learning inference on IoT." ACM Transactions on Embedded Computing Systems (TECS), 18, no. 5s: 1-22 (2019).

CONFERENCE PAPERS

- [C5] **Ching-Yi Lin**, and Sahil Shah. "Low-Bit Integerization of Vision Transformers using Operand Reordering for Efficient Hardware." 2025 IEEE 67th International Midwest Symposium on Circuits and Systems (MWSCAS).
- [C4] **Ching-Yi Lin**, Yann Gilpin, Zixin Chen, Elizabeth Wayne and Marc Dandin. "Towards a Lab-on-a-Chip as a Service (LoCaaS) Framework for in Vitro Cancer Cell Assays." 2024 22nd IEEE Interregional NEWCAS Conference (NEWCAS), pp. 268-272, doi: 10.1109/NewCAS58973.2024.10666355 (2024).
- [C3] Yann Gilpin, **Ching-Yi Lin**, Mats Forssell, Siyang Zheng, Pulkit Grover, and Marc Dandin. "Tracking the effects of tumor treating fields on human breast cancer cells in vitro using a capacitance sensing lab-on-CMOS microsystem." 2022 29th IEEE International Conference on Electronics, Circuits and Systems (ICECS), pp. 1-4 (2022).
- [C2] Yuwei Qin, Ruben Purdy, Alec Probst, **Ching-Yi Lin**, and Jian-Gang Jimmy Zhu. "Non-linear CNN-based read channel for hard disk drive with 30% error rate reduction and sequential 200Mbits/second throughput in 28nm CMOS." 2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits), pp. 206-207 (2022).
- [C1] **Ching-Yi Lin**, and Radu Marculescu. "Model personalization for human activity recognition." 2020 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops), pp. 1-7 (2020).

PREPRINTS AND PENDING PAPERS

- [PP2] **Ching-Yi Lin**, and Sahil Shah. "Systolic Array-based Architecture for Low-Bit Integerized Vision Transformers." arXiv preprint arXiv:2508.20334 (2025) (Submitted to HPCA 2025).
- [PP1] Irem Didin, Carl Brando, **Ching-Yi Lin**, and Sahil Shah. "Characterization and Modeling of Multilevel Analog ReRAM Synapses in the Sky130 Process." Authorea Preprints (2025) (Submitted to IEEE Journal of Exploratory Solid-State Computational Devices and Circuits).

PATENTS

- [P1] Marc Dandin, and **Ching-Yi Lin**. "System and Method to Measure CAR-T cell quality," U.S. Patent #0366873, November 2023

COMPETITIONS

IEEE Signal Processing Cup Hardware Designer

March 2017

- *Excellent video and entertaining concept* Award
- Implemented the signal processing algorithm into embedded system
- Responsible for system design (Raspberry + Arduino), task parallelization, and protocol design.

Eurobot 2016, International Robotics Competition Software Team Leader

June 2016

- Led software team of DIT Robotics, including 9 mechanical engineering students, 1 physic student and 1 EE student.
- Responsible for embedded system design and communication protocol design.

OUTREACH ACTIVITIES

Session Committee at Qual Practice Sessions from CMU ECE Graduate Organization	2024
Graduate Student Spotlight , CMU Office of Graduate and Postdoctoral Affairs	2024
Speaker at Student Panel of CMU New Graduate Student Orientation	2024
Panelist in CMU ECE Graduate Student Panel	2021
Panelist in CMU ECE Open House	2020
Student Contact in Prospective Student's Admission Letter Eunseok Lee, Eric Tang, Alhad Daftardar, Yitu Wang, and Luke Sammarone	2020
Excellent video and entertaining concept Award IEEE Signal Processing Cup	2017

OTHER ACTIVITIES

Taiwanese Scholar Society · President in 2019-2020 · Vice president in 2020-2022	2019-2022
Team Leader , NTHU EE Tennis Team	2014-2015

REFERENCES

Pamela A. Abshire Chair and Professor Electrical and Computer Engineering Stony Brook University	Sahil Shah Assistant Professor Electrical and Computer Engineering University of Maryland, College Park	Marc Dandin Associate Professor Electrical and Computer Engineering Carnegie Mellon University
L. Richard Carley Professor Electrical and Computer Engineering Carnegie Mellon University	Carlee Joe-Wong Professor Electrical and Computer Engineering Carnegie Mellon University	

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